Oscillator

Q. An oscillator converts ……………..

1. dc. power into d.c. power
2. dc. power into a.c. power
3. mechanical power into a.c. power
4. none of the above

Answer : 2

Q. In an LC transistor oscillator, the active device is ……………

1. LC tank circuit
2. Biasing circuit
3. Transistor
4. None of the above

Answer : 3

Q. In an LC oscillator, the frequency of oscillator is …………… L or C.

1. Proportional to square of
2. Directly proportional to
3. Independent of the values of
4. Inversely proportional to square root of

Answer : 4

Q An oscillator produces……………. oscillations

1. Damped
2. Undamped
3. Modulated
4. None of the above

Answer : 2

Q. An oscillator employs ……………… feedback

1. Positive
2. Negative
3. Neither positive nor negative
4. Data insufficient

Answer : 1

Q. An LC oscillator cannot be used to produce ………………. frequencies

1. High
2. Audio
3. Very low
4. Very high

Answer : 3

Q. Hartley oscillator is commonly used in ………………

1. Radio receivers
2. Radio transmitters
3. TV receivers
4. None of the above

Answer : 1

Q. In a phase shift oscillator, we use …………. RC sections

1. Two
2. Three
3. Four
4. None of the above

Answer : 2

Q. In a phase shift oscillator, the frequency determining elements are ………..

1. L and C
2. R, L and C
3. R and C
4. None of the above

Answer : 3

Q. A Wien bridge oscillator uses …………… feedback

1. Only positive
2. Only negative
3. Both positive and negative
4. None of the above

Answer : 3

Q. The piezoelectric effect in a crystal is …………..

1. A voltage developed because of mechanical stress
2. A change in resistance because of temperature
3. A change in frequency because of temperature
4. None of the above

Answer : 1

Q. The crystal oscillator frequency is very stable due to ……………… of the crystal

1. Rigidity

2. Vibrations
3. Low Q
4. High Q

Answer : 4

Q. The application where one would most likely find a crystal oscillator is …………….

1. Radio receiver
2. Radio transmitter
3. AF sweep generator
4. None of the above

Answer : 2

Q. An oscillator differs from an amplifier because it ………

1. Has more gain
2. Requires no input signal
3. Requires no d.c. supply
4. Always has the same input

Answer : 2

Q. One condition for oscillation is …………

1. A phase shift around the feedback loop of 180°
2. A gain around the feedback loop of one-third
3. A phase shift around the feedback loop of 0°
4. A gain around the feedback loop of less than 1

Answer : 3

Q. A second condition for oscillations is ………………

1. A gain of 1 around the feedback loop
2. No gain around the feedback loop
3. The attention of the feedback circuit must be one-third
4. The feedback circuit must be capacitive

Answer : 1

Q In a certain oscillator A_v = 50. The attention of the feedback circuit must be ………..

1. 1
2. 0.1
3. 10
4. 0.2

Answer : 4

Q. For an oscillator to properly start, the gain around the feedback loop must initially be

1. 1
2. Greater than 1
3. Less than 1
4. Equal to attenuation of feedback circuit

Answer : 2

Q. In Colpitt's oscillator, feedback is obtained …………….

1. By magnetic induction
2. From the centre of split Inductors
3. From the centre of split capacitors
4. None of the above

Answer : 3

Q. The Q of the crystal is of the order of ………..

1. 100
2. 1000
3. 50
4. More than 10,000

Answer : 4

Q. Quartz crystal is most commonly used in crystal oscillators because …………

1. It has superior electrical properties
2. It is easily available
3. It is quite inexpensive
4. None of the above

Answer : 1

Q. In an LC oscillator, if the value of L is increased four times, the frequency of oscillations is ………..

1. Increased 2 times
2. Decreased 4 times
3. Increased 4 times
4. Decreased 2 times

Answer : 4

Q. An important limitation of a crystal oscillator is ……………

1. Its low output

2. Its high Q
3. Less availability of quartz crystal
4. Its high output

Answer : 1

Q. The signal generator generally used in the laboratories is …………. oscillator

1. Wien-bridge
2. Hartely
3. Crystal
4. Phase shift

Answer : 1
